

Spring 2026 Characterization Report

Group: __3__

Members:

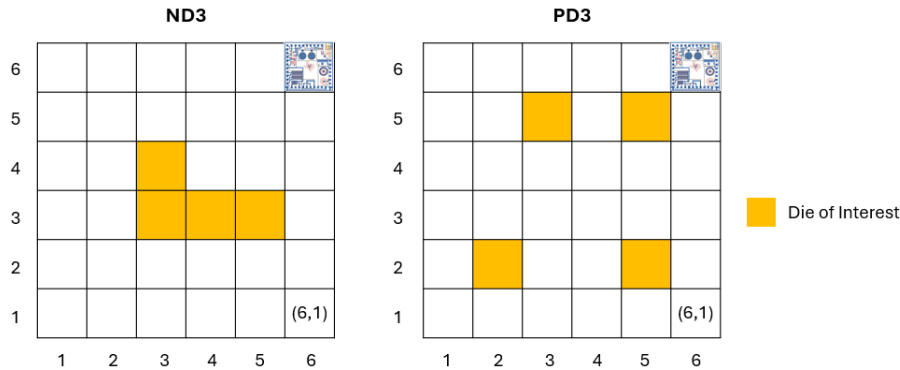
- **Wei-Cheng Lee**
- **Grant Reeder**
- **Jimena Casas**
- **Adrian Corrales**
- **Zander Hubbard**

Notes:

- Submit your .csv files from measurement as part of the Canvas assignment.
- Please complete this report as a group. Add your group number and names above.
- Refer to the lab manual on Canvas for more details on characterization. This report is just a skeleton of the lab manual. Feel free to add more space as needed.
- It's expected for many of your devices to not work correctly or at all. However, this is not a reason to not report your results or provide explanations for any failures. Always attempt to measure additional devices (feel free to use any sample you can find including the TA samples) and, if all else fails, provide an explanation for why your devices do not work. For example, "fab was bad" is not an acceptable explanation. List specific fabrication steps that may have caused your devices to fail, and why. Explain what your results should be for a working device.
- The parts included in this report are the minimum. Feel free to perform additional characterization to try for up to 20 points extra credit (without breaking the probe stations, please).
- Always include units.

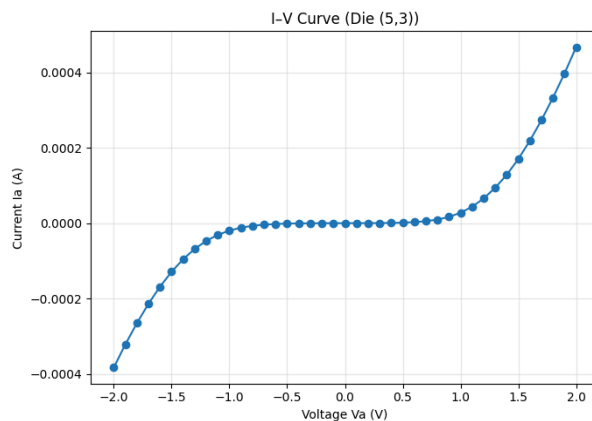
Part 1: Resistors

Die Map:



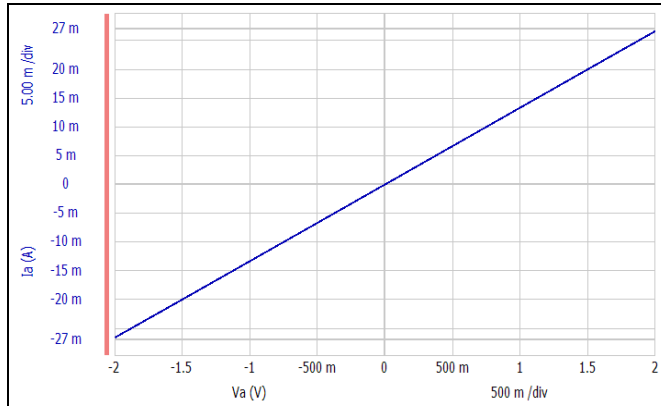
I-V characteristics

- For both n- and p-type samples, include plots of the I-V characteristics of one good R2 resistor. (2 pts)
- n-type sample:*



Although the resistor was probed correctly and the substrate was left unbiased, the nonlinear I-V curve suggests non-ohmic metal–semiconductor contact behavior, likely caused by oxide-related fabrication issues. This is consistent with the abnormal sheet resistance observed after gate oxidation, where measurable resistance was obtained even though the oxide should have been insulating, suggesting incomplete or non-uniform oxide formation or imperfect contact-window etching. Residual oxide in the contact region could degrade the Al-to-diffusion contact, introducing a contact barrier and large parasitic contact resistance. This likely explains both the nonlinear I-V behavior and the measured resistance being large and various.

p-type sample:



Resistance

■ Measure the resistance of 4 good resistors of each type on each substrate. (6 pts)

n-type sample:

		Resistance (Ω)	Location
R1	1	129.87	(5, 3)
	2	121.36	(3,4)
	3	111.11	(3,3)
	4	129.62	(4,3)
	Average	123.00	-
R2	1	161.03	(5, 3)
	2	166.53	(3,4)
	3	155.76	(3,3)
	4	222.97	(4,3)
	Average	176.57	-
R3	1	8620	(5, 3)
	2	7540	(3,4)
	3	1390	(3,3)
	4	7700	(4,3)

	Average	6312.5	-
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■ Do the resistances vary across your sample? What might cause the variation, if any? (1 pt)

Resistor	Average (Ω)	Largest Deviation	% Deviation
R1	123	12 Ω	9.8%
R2	176.6	46.4 Ω	26.3%
R3	6310	4920 Ω	78.0%

Resistance variation across the wafer is mainly attributed to **process non-uniformity**, including spatial variations in dopant diffusion, sheet resistance, contact resistance, and lithographic dimensions. Variations in diffusion concentration or junction depth change the sheet resistance R_s , while differences in contact quality and alignment contribute additional resistance variation. In particular, slight **mask misalignment** observed on the ND3 sample after each lithography step may have contributed to changes in resistor geometry or contact overlap, further increasing resistance variability. For the serpentine resistor R3, bend resistance and current crowding further amplify these effects, making its variation much larger than R1 and R2. Local fabrication defects may also contribute to outliers.

Possible Reasons:

1. Diffusion non-uniformity

The manual emphasizes resistor values depend on sheet resistance R_s set by doping diffusion.

Variations in:

- Dopant concentration (pre-deposition nonuniformity)
- Junction depth / drive-in diffusion

can change R_s across different die locations.

2. Lithography defects

Actual resistor width may differ from designed width due to:

- Over/under PR develop
- Over/under etching
- Misalignment

Since resistance is inversely proportional to width, narrower regions increase resistance.

p-type sample:

		Resistance	Location
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R1	1	44.38	(2,2)
	2	45.50	(3,5)
	3	47.13	(5,5)
	4	45.20	(5,2)
	Average	45.55	-
R2	1	74.23	(2,2)
	2	76.20	(3,5)
	3	76.35	(5,5)
	4	75.26	(5,2)
	Average	75.51	-
R3	1	228.26	(2,2)
	2	232.06	(3,5)
	3	239.25	(5,5)
	4	246.28	(5,2)
	Average	236.46	-

■ Do the resistances vary across your sample? What might cause the variation, if any? (1 pt)

The resistances across the p-type sample show relatively low variation compared to the n-type sample, indicating better process uniformity.

Small variations can still be attributed to:

- Slight differences in dopant concentration across the wafer
- Lithography variations affecting resistor width and length
- Contact resistance differences at probe points

Because resistance is inversely proportional to cross-sectional area, even small dimensional variations from etching or development can introduce measurable differences. However, the tighter grouping of values suggests more consistent fabrication for the p-type devices.

■ Using the average values from the tables above, calculate the following values. Include your work. (9 pts)

	n-type sample	p-type sample
Sheet resistance (R_s)	5.36 $\Omega/\square$	2.996 $\Omega/\square$
Contact resistance (R_c)	34.72 Ω	15.55 Ω
Effective resistance of bends (R_{BND})	480.14 Ω	5.63 Ω
Effective size of a bend (L_{BND})	89.6 $\square$	1.88 $\square$

The sheet resistance and contact resistance could be expressed by the following two equations:

$$R_s = \frac{R_2 - R_1}{N_2 - N_1}, \quad R_c = \frac{R_2 - N_2 R_s}{2},$$

where $N_1 = 250 / 25 = 10$ and $N_2 = 500 / 25 = 20$.

Based on Fig. 1, we estimated the lengths of the bend resistor and body resistor in R3 to be approximately $1.25 \times R_1$ and $0.75 \times R_1$, respectively. Accordingly, we obtain $N_3 = 6 \times 0.75 N_1$ and $N_{BND} = 1.25 N_1$. Using these values, we can apply the corresponding equations to calculate the effective bend resistance (R_{BND}) and the effective bend size (L_{BND}).

$$R_{BND} = \frac{R_3 - 2R_c - N_3 R_s}{N_B}, \quad L_{BND} = \frac{R_{BND}}{R_s}$$

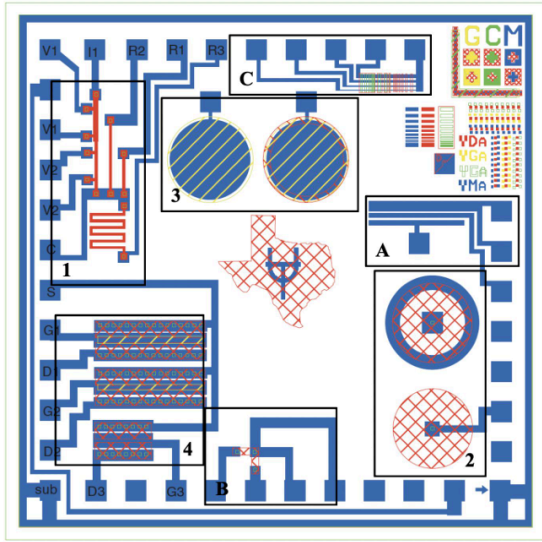
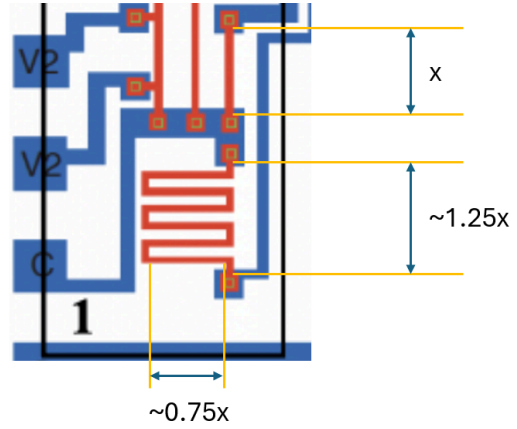


Figure 1: Composite Drawing of Holberg Mask Set.

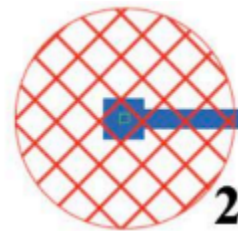


Part 2: Diodes

For the following, refer to your diodes as:

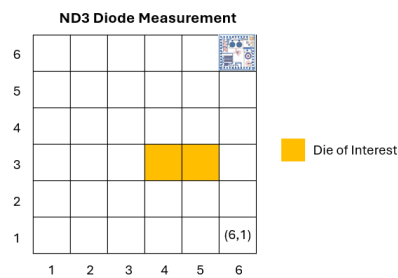


type 1 (with field relief plate)



type 2

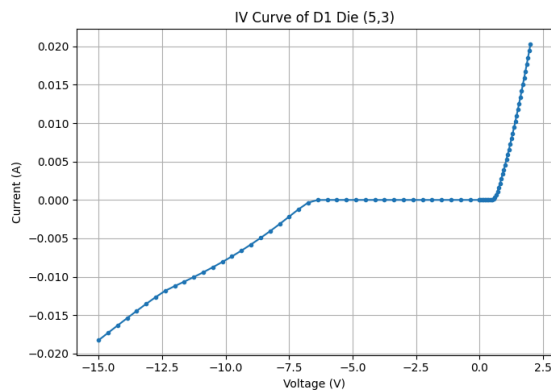
Die map



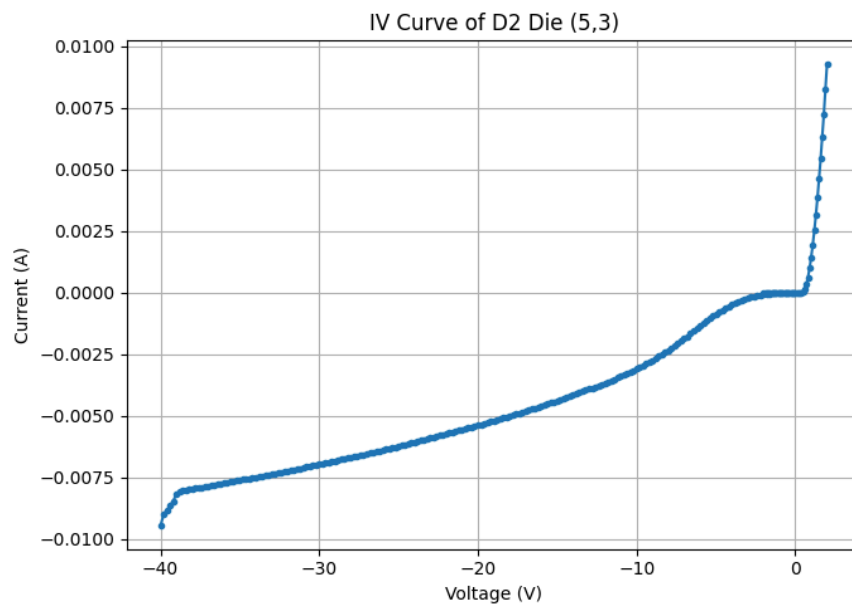
I-V characteristics

■ For both your n- and p-type samples, include plots of the I-V characteristics of one good diode of each type. These must show the I-V characteristics under forward and reverse bias. (4 pts)

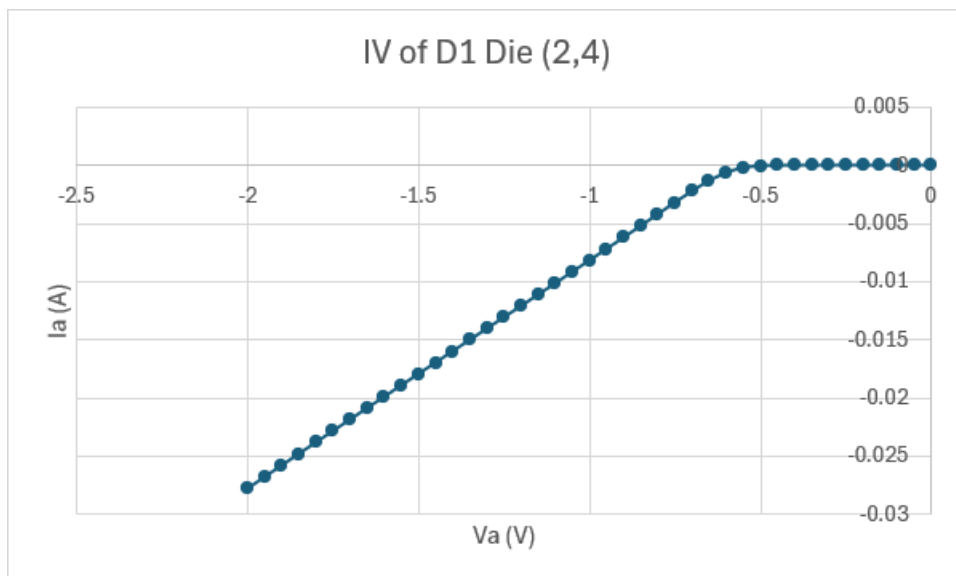
n-type sample, type 1:



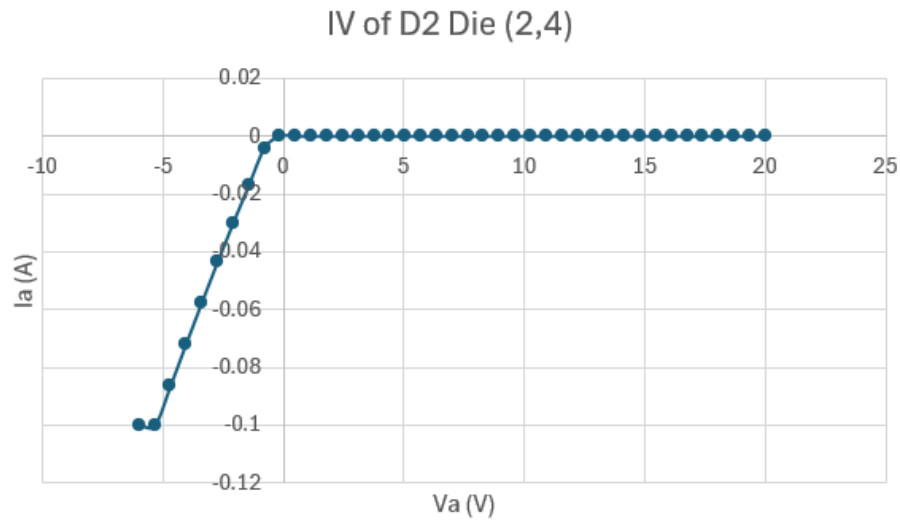
n-type sample, type 2:



p-type sample, type 1:



p-type sample, type 2:



■ Fill in the table below. (8 pts)

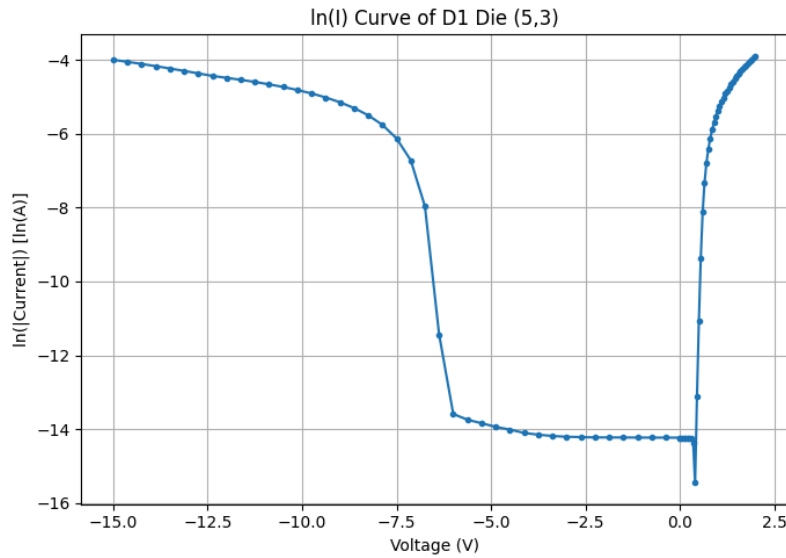
	n-type sample		p-type sample	
	Type 1	Type 2	Type 1	Type 2
Forward resistance (R_F)	57.41 Ω	104.21 Ω	51.03 Ω	50.19 Ω
Breakdown voltage (V_{BR})	-6.3 V	-0.8V	Not Observed	Not Observed
Reverse leakage current (I_L)	0.68 μA	0.81 μA	2.89 μA	4.77 μA
Turn on voltage (V_O)	0.84 V	~1.04V	Not Observed	Not Observed

The turn-on voltage V_O and breakdown voltage V_{BR} were not clearly observed in the p-type diode measurements due to limitations in the applied voltage sweep range and the measurement conditions. For forward bias, the voltage range did not extend sufficiently into the high-current exponential region needed to clearly define a sharp turn-on point, and the presence of series resistance further smoothed the transition, making V_O less distinct. In reverse bias, the sweep did not reach a high enough voltage magnitude to induce avalanche or Zener breakdown, particularly for the Type 1 diode, which is designed with a field plate to increase breakdown voltage. Additionally, relatively high leakage current and

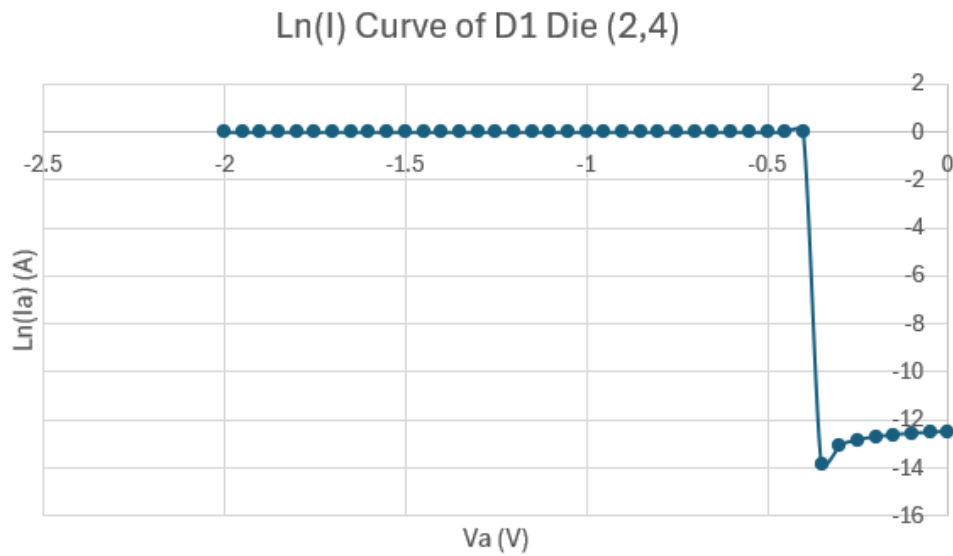
non-ideal effects such as interface traps and contact resistance can obscure the onset of breakdown, causing the I–V curve to appear gradual rather than exhibiting a sharp increase. As a result, both V_O and V_{BR} could not be definitively extracted from the measured data.

■ Plot your I-V data on a semi-log scale (i.e., $\ln(I)$ vs. V) and fit the diode I-V to the ideal diode equation for one diode of any kind on both substrates. Include your plots and fill in the table below. State which diode type you chose. (6 pts)

n-type sample, type __1__:



p-type sample, type __1__:



	n-type sample	p-type sample
Ideality factor (n)	1.16	1.47
Dark saturation current (I_0)	7.3×10^{-13} A	9.83×10^{-11} A

Equations used to calculate n and I_0

$$\ln(I) = \ln(I_S) + \frac{qV}{nkT}$$

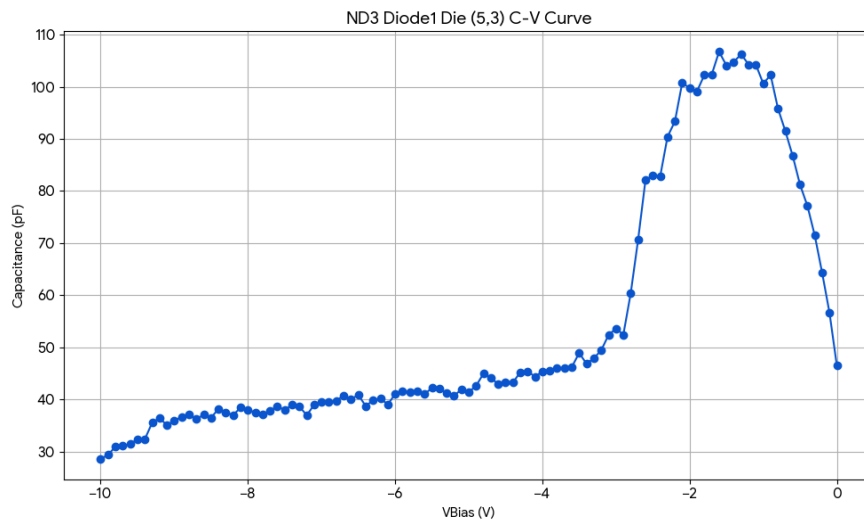
$$n = \frac{q}{mkT}$$

, where m is the slope of the forward region.

C-V characteristics

■ For both your n- and p-type samples, include plots of the C-V characteristics of one good diode of any kind. State which diode type you chose. (2 pts)

n-type sample, type __1__:



Ideally, the junction capacitance should increase monotonically as the reverse bias approaches 0 V due to the shrinking depletion width. However, an abnormal drop in capacitance is observed near 0 V in this measurement.

Possible causes include:

1. **Junction damage or defects introduced during diffusion:**

Increase interface trap density and leakage.

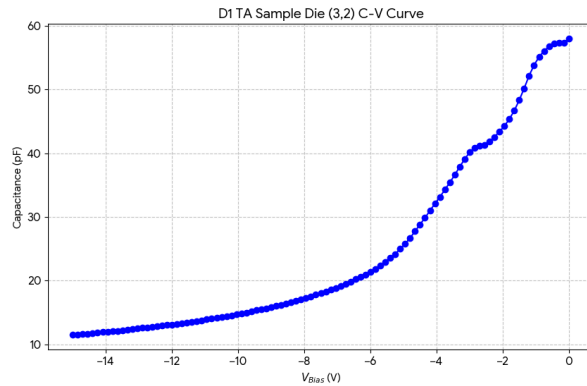
2. **Non-uniform oxide growth:**

Introduce additional interface states.

3. **Lithography misalignment:**

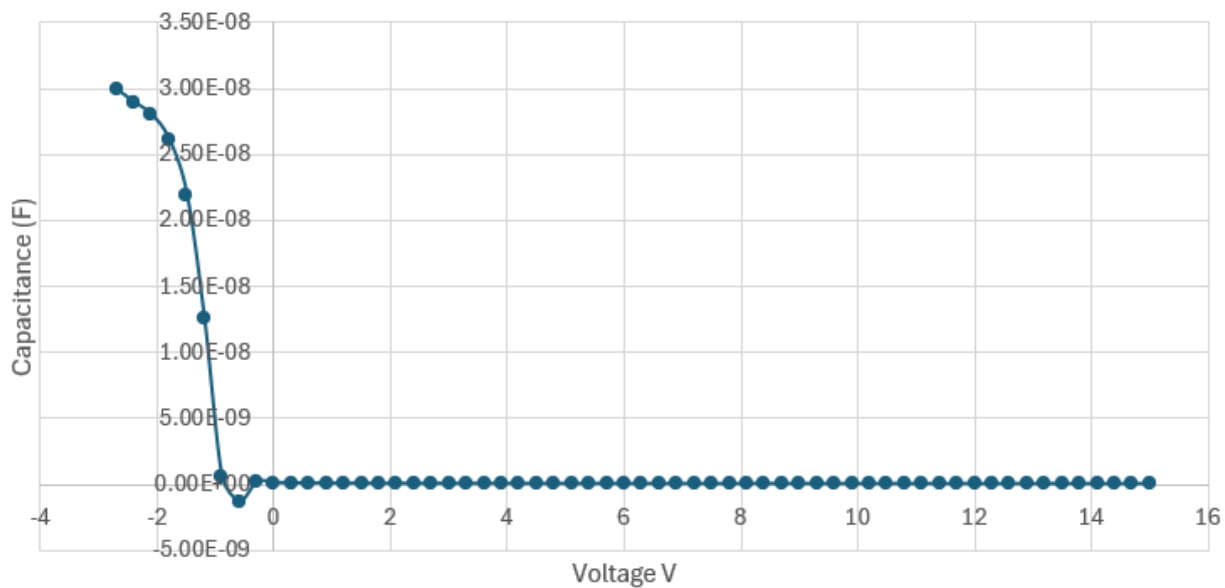
Introduce parasitic resistance.

Expected C-V curve might look like:



p-type sample, type __1__:

Type 1 Diode 1 kHz C-V (2,4)



The p-type diode C–V curve does not exhibit the ideal monotonic increase toward 0 V, suggesting non-ideal junction behavior.

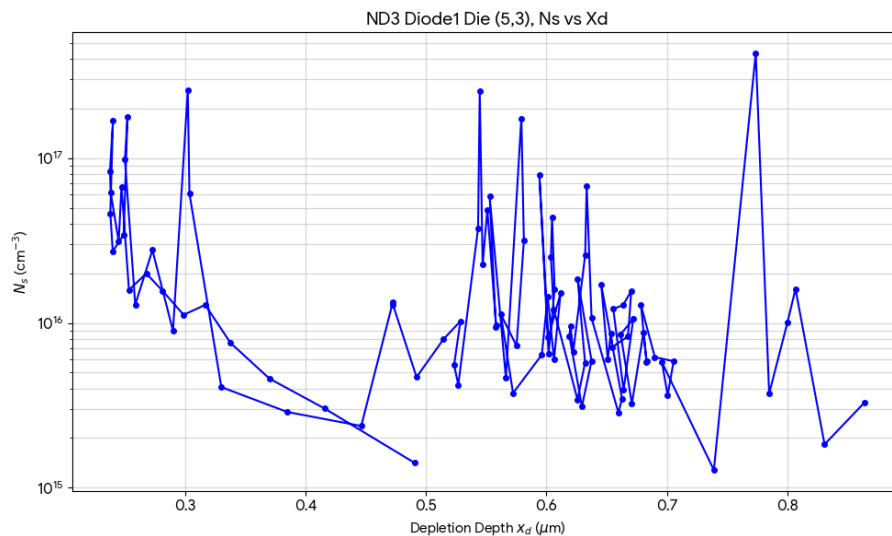
Possible causes include:

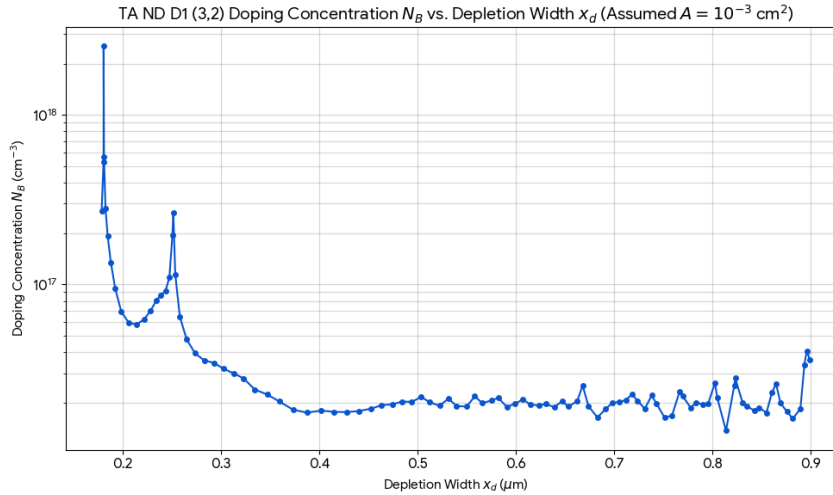
- High interface trap density affecting capacitance measurement
- Series resistance limiting accurate measurement at low bias
- Measurement noise or probe instability

Ideally, capacitance should increase as reverse bias decreases due to shrinking depletion width, but deviations indicate fabrication or measurement limitations.

■ Using the C-V characteristics obtained above, calculate and plot the approximate background doping concentration N_B as a function of x_d . (3 pts)

n-type sample, type __1__:





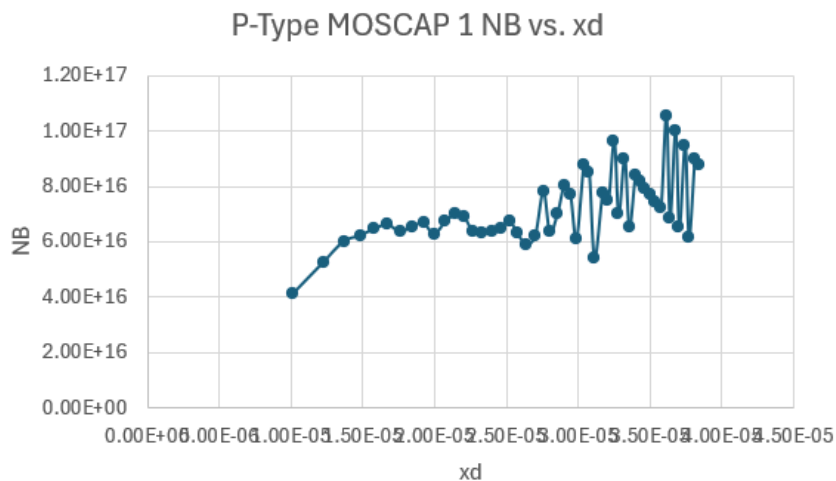
For the depletion width (x_d), it's calculated as:

$$x_d(V_{RB}) \approx \frac{A\epsilon_s}{C(V_{RB})}$$

The doping concentration at the edge of the depletion region is inversely proportional to the slope of the $1/C^2$ versus V curve. The formula used is:

$$N_B(V_{RB}) \approx -2 \left(q\epsilon_s A^2 \left(\frac{d\left(\frac{1}{C^2}\right)}{dV_{RB}} \right) \right)^{-1}$$

p-type sample, type 1:



The extracted doping profile from the diode C-V data shows a smooth and physically reasonable trend. At smaller depletion widths, the doping concentration is lower, around $4 \times 10^{16} \text{ cm}^{-3}$, which corresponds

to the region near the junction where the doping profile is transitioning. As the depletion width increases, the doping concentration stabilizes in the range of approximately $6 \times 10^{16} \text{ cm}^{-3}$, indicating a more uniform bulk substrate doping level.

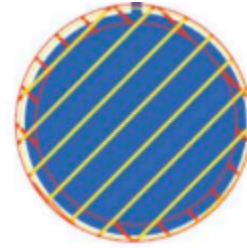
The gradual change in doping concentration reflects the expected transition from the junction region into the bulk material. While some variation is observed at larger depletion widths, this is likely due to the sensitivity of the numerical differentiation used in the extraction process, as well as measurement noise in the capacitance data. Overall, the profile is consistent with a reasonably uniform substrate doping and demonstrates that the diode C–V method provides a reliable estimate of the doping concentration.

Part 3: MOSCAPs

For the following, refer to your MOSCAPs as:

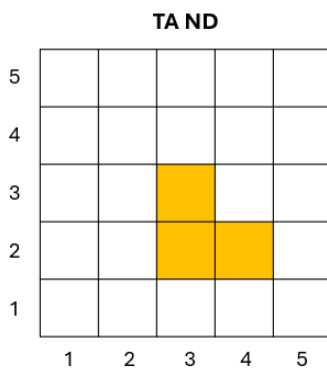


type 1



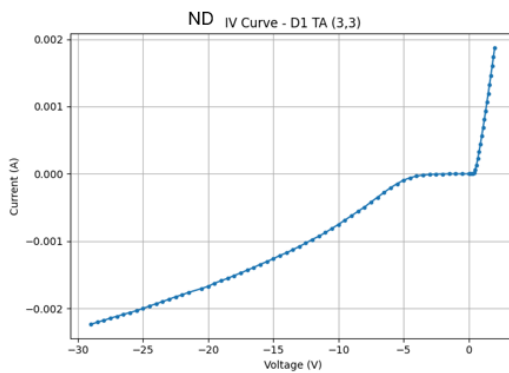
type 2 (with diffused guard ring)

Die Map



I-V characteristics

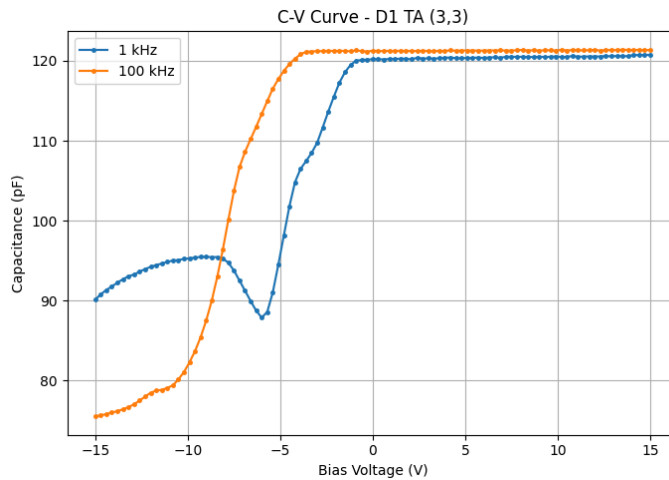
■ Find a few good MOSCAPs on each substrate, preferably in several locations. Use these for C-V measurements.



C-V characteristics

■ For both your n- and p-type samples, include plots of the C-V characteristics of one good MOSCAP of any kind. State which MOSCAP type you chose. (2 pts)

n-type sample, type __1__:

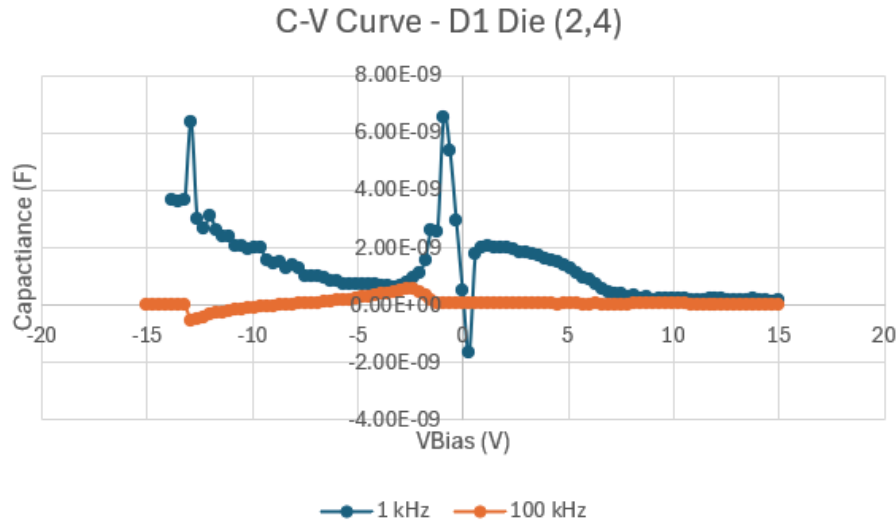


Our device did not produce reliable or usable C-V data, so the measurement shown here is based on the TA's sample. We believe the discrepancy between our device and the TA's sample is mainly due to alignment errors during fabrication and differences in the oxide growth process, which likely degraded the quality of our device.

Compared to an ideal MOSCAP C-V curve, this result still shows non-ideal behavior. Ideally, the capacitance should be flat in accumulation, decrease smoothly in depletion, and saturate in inversion with a sharp transition and minimal frequency dependence. However, this curve is not flat in accumulation, shows a dip around -6 V, and exhibits noticeable frequency dispersion between 1 kHz and 100 kHz.

These non-idealities are likely caused by interface trap states, which lead to frequency dispersion and stretch-out, along with oxide charges that distort the curve. Additional factors such as series resistance, parasitic capacitance, and minor oxide defects or leakage paths may also contribute to the abnormal dip and overall deviation from the ideal behavior.

p-type sample, type __1__:



- For both your *n*- and *p*-type samples, find the following data for 3 good MOSCAPs of each kind. (8 pts)

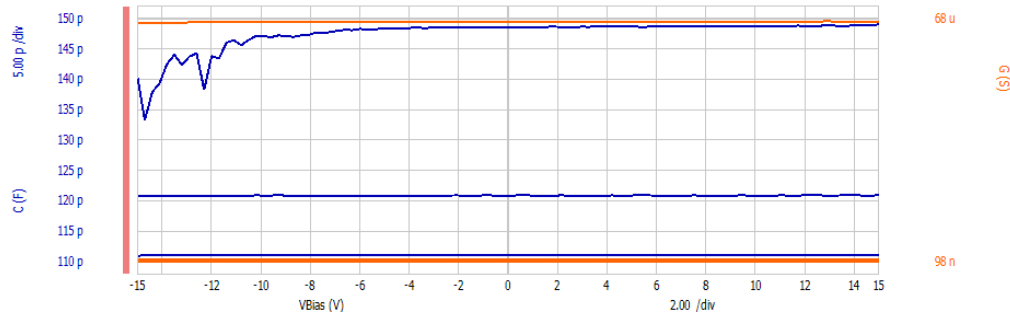
n-type sample:

		Threshold voltage (V_{TH})	Flat band voltage (V_{FB})
Type 1	1 (3,3)	-4.5 V	-2.4 V
	2 (3,2)	-4.3 V	-2.2 V
	3 (4,2)	-4.2 V	-2 V
Type 2	1	x (Unmeasurable)	x
	2	x	x
	3	x	x

- Comment on the device-to-device variability of these parameters. (1 pt)

For the type 1 MOSCAP, the C-V curves look good, and the extracted V_{TH} and V_{FB} values from three different dies are close to each other.

C-V Curve for the Type 2 MOSCAP:



For the type 2 MOSCAP, the flat C - V curve likely indicates that the measured capacitance is dominated by parasitic capacitance or leakage, rather than the MOS capacitor itself. A likely process-flow issue is guard-ring misalignment or unintended electrical connection. If the guard ring overlaps incorrectly, **shorts to the gate, substrate, or metal pad, or creates a leakage path** around the capacitor, the applied bias will not properly modulate the MOS depletion region. As a result, the measured capacitance stays nearly constant instead of showing a normal accumulation–depletion–inversion transition.

p-type sample:

		Threshold voltage (V_{TH})	Flat band voltage (V_{FB})
Type 1	1	0.9 V	-1.8 V
	2	1.1 V	-2.0 V
	3	1.0 V	-1.7 V
Type 2	1	1.8 V	-1.8 V
	2	2.1 V	-1.9 V
	3	1.1 V	-1.6 V

■ Comment on the device-to-device variability of these parameters. (1 pt)

The p-type MOSCAPs show relatively low device-to-device variability, especially for the Type 1 devices. For Type 1, V_{FB} only varies from about -2.0 V to -1.7 V, while V_{TH} varies from about 0.9 V to 1.1 V. This small spread suggests that the oxide thickness, oxide charge, and substrate doping are fairly consistent across those measured devices. The Type 2 MOSCAPs show slightly more variation, with V_{TH} ranging from about 1.8 V to 2.1 V, while V_{FB} remains close to -1.8 V. The larger Type 2 threshold spread is likely due to the guard ring structure introducing additional parasitic capacitance, leakage paths, or sensitivity to alignment errors. Overall, the flatband voltages are

fairly consistent, while the threshold voltages show more variability, especially in the Type 2 MOSCAPs.

■ For both your n- and p-type samples, find the following data for one good MOSCAP of each kind. Include your work. (6 pts)

	n-type sample		p-type sample	
	Type 1	Type 2	Type 1	Type 2
Gate-oxide capacitance per unit area (C_{ox})	121 pF	150 pF	350 pF	323 pF
Gate-oxide thickness (t_{ox})	68 nm	65 nm	15.7 nm	30.2 nm

n-type 1:

$$C_{ox} \approx C_{max} = 120 \text{ pF.}$$

$$t_{ox} = \frac{\epsilon_{ox} A}{C_{ox}}, \text{ where } A = \pi(225 \times 10^{-6})^2 \rightarrow t_{ox} = \frac{\epsilon_{ox} A}{C_{ox}} = \frac{3.9\epsilon_o A}{C_{ox}} \approx 68nm.$$

n-type 2:

$$C_{ox} \approx C_{max} = 150 \text{ pF.}$$

$$t_{ox} = \frac{\epsilon_{ox} A}{C_{ox}}, \text{ where } A = \pi(300 \times 10^{-6})^2 \rightarrow t_{ox} = \frac{\epsilon_{ox} A}{C_{ox}} = \frac{3.9\epsilon_o A}{C_{ox}} \approx 65nm.$$

p-type 1:

$$C_{ox} \approx C_{max} = 350 \text{ pF}$$

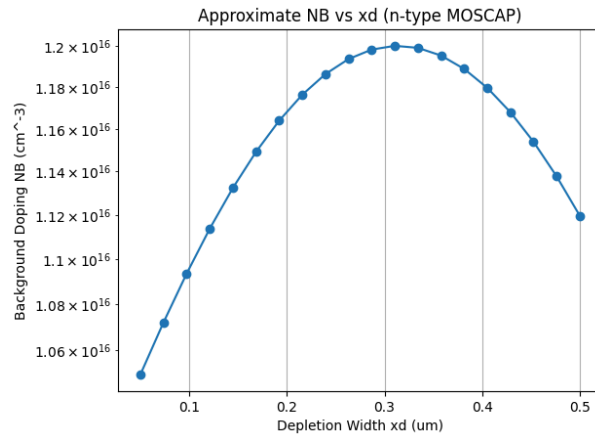
$$t_{ox} = \frac{\epsilon_{ox} A}{C_{ox}}, \text{ where } A = \pi(225 \times 10^{-6})^2 \rightarrow t_{ox} = \frac{\epsilon_{ox} A}{C_{ox}} = \frac{3.9\epsilon_o A}{C_{ox}} \approx 15.7nm.$$

p-type 2:

$$C_{ox} \approx C_{max} = 323 \text{ pF}$$

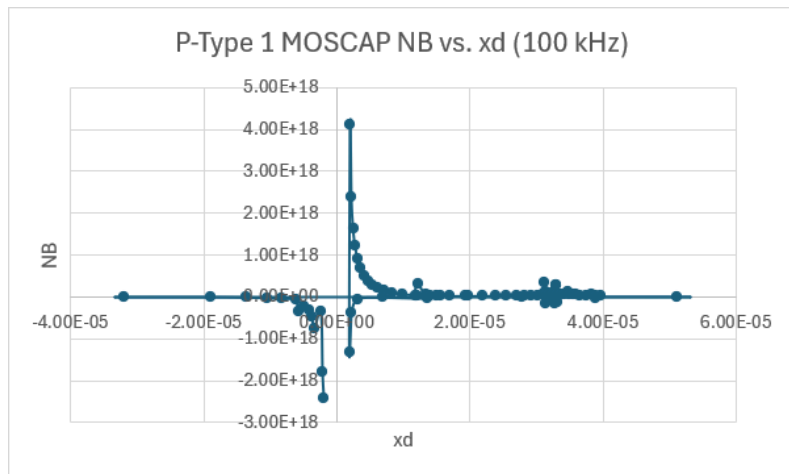
$$t_{ox} = \frac{\epsilon_{ox} A}{C_{ox}}, \text{ where } A = \pi(300 \times 10^{-6})^2 \rightarrow t_{ox} = \frac{\epsilon_{ox} A}{C_{ox}} = \frac{3.9\epsilon_0 A}{C_{ox}} \approx 30.2 \text{ nm}.$$

n-type sample: Calculate and plot the approximate background doping concentration N_B as a function of x_d .



(Same approach as the diode NB vs. x_d analysis.)

p-type sample: Calculate and plot the approximate background doping concentration N_B as a function of x_d .



The p-type Type 1 MOSCAP doping profile shows large fluctuations and several non-physical negative values. This is mainly due to the derivative-based extraction method, where N_B depends on the inverse slope of the $1/C^2$ versus voltage curve. Small noise in the measured capacitance is amplified when taking the numerical derivative, producing spikes and sign changes in the extracted doping concentration. The

instability is especially noticeable near small depletion widths, where small changes in capacitance can cause very large changes in the calculated slope. Negative doping values are not physically meaningful and indicate measurement noise, parasitic capacitance, interface trap effects, or numerical error in the derivative. The more stable region at larger depletion widths is the most useful for estimating the background doping concentration, which appears to be on the order of 10^{15} to 10^{16} cm^{-3} . Overall, the MOSCAP extraction gives a noisy but reasonable estimate of the substrate background doping.

■ How do your gate-oxide thicknesses compare to the ones measured after gate-oxidation? (1 pt)

For the **n-type** device, the oxide thickness measured after gate oxidation is approximately 17 nm, while the thickness extracted from the C-V data is around 65 nm. This discrepancy is primarily due to the strong dependence of the extracted thickness on the assumed oxide capacitance C_{ox} . For example, using $C_{ox} = 150 \text{ pF}$ for a $600 \text{ }\mu\text{m}$ diameter capacitor leads to an estimated thickness of about 65 nm. In contrast, the physically measured thickness of 17 nm corresponds to an oxide capacitance of approximately 570 pF.

The key issue is that the **measured capacitance is significantly smaller** than expected, indicating that it does not represent the true oxide capacitance. A likely cause is related to the type 2 device structure with a guard ring. If the guard ring is misaligned, floating, shorted, or improperly biased, **a portion of the intended capacitor area may not effectively contribute to the measurement**. As a result, the effective capacitance is reduced.

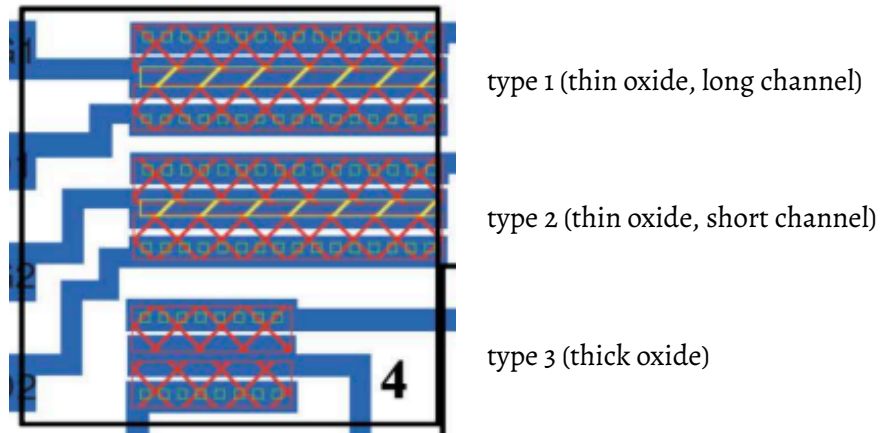
Because the extracted oxide thickness is inversely proportional to capacitance, this reduced effective capacitance leads to an artificially larger calculated thickness. In other words, the discrepancy is not due to actual oxide growth variation, but rather to measurement artifacts arising from process or connection issues in the guard ring structure.

■ Compare the doping concentrations to the ones extracted using diode C-V. (2 pts)

For the n-type device, the diode-extracted $N_s(x_d)$ shows strong variation near the surface, with a sharp peak around 10^{18} cm^{-3} , then gradually decreases and becomes relatively flat ($\sim 10^{15} - 10^{16} \text{ cm}^{-3}$) at larger x_d . This reflects the junction profile, where doping changes rapidly near the surface and transitions to the bulk substrate. In contrast, the MOSCAP result is generally flatter since it mainly measures the background substrate doping. The two methods are consistent in the deeper region ($x_d > 0.4 \text{ }\mu\text{m}$), where both indicate similar bulk doping levels, while the diode appears noisier due to derivative-based extraction and higher sensitivity to measurement variation.

Part 4: MOSFETs

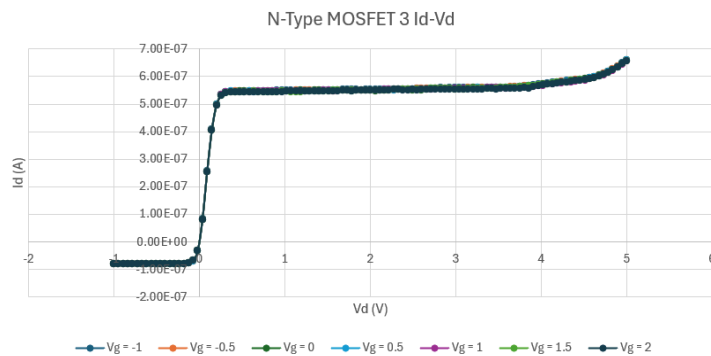
For the following, refer to your MOSFETs as:



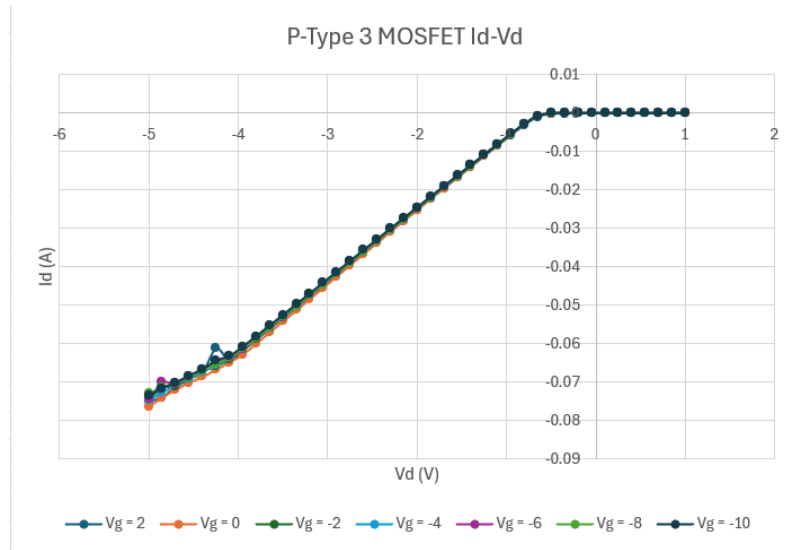
I-V characteristics

■ For both your n- and p-type samples, include coarse sweep plots of the I_D - V_D characteristics of one good MOSFET of any kind. State which MOSFET you chose. (2 pts)

n-type sample, type __3__:



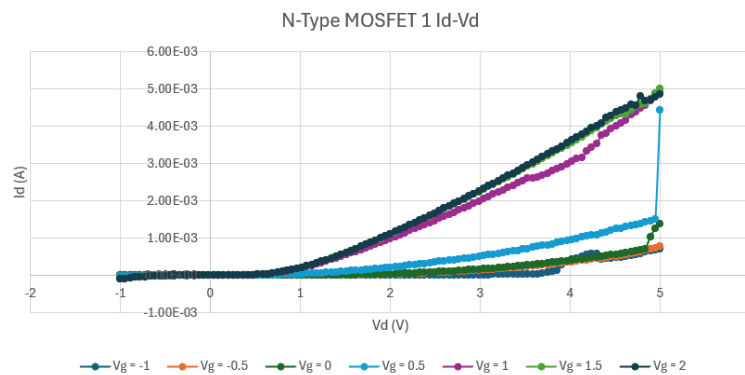
p-type sample, type __3__:



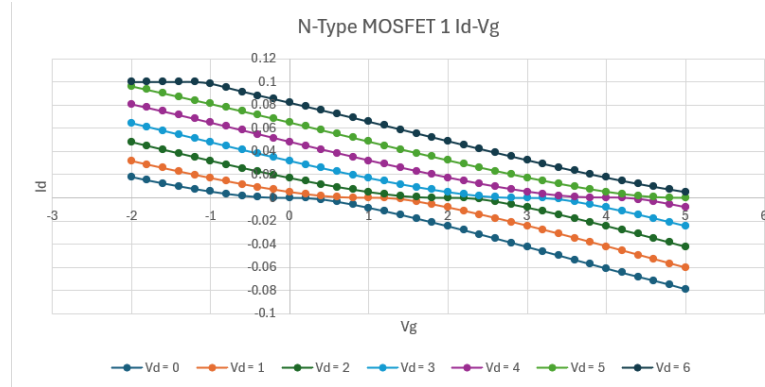
■ For both your n- and p-type samples, include fine sweep plots of the I_D - V_D and I_D - V_G family of curves for one good MOSFET of each of the 3 kinds. Using the I_D - V_D characteristics, plot the transconductance g_m as a function of gate voltage. (18 pts)

n-type sample, type 1:

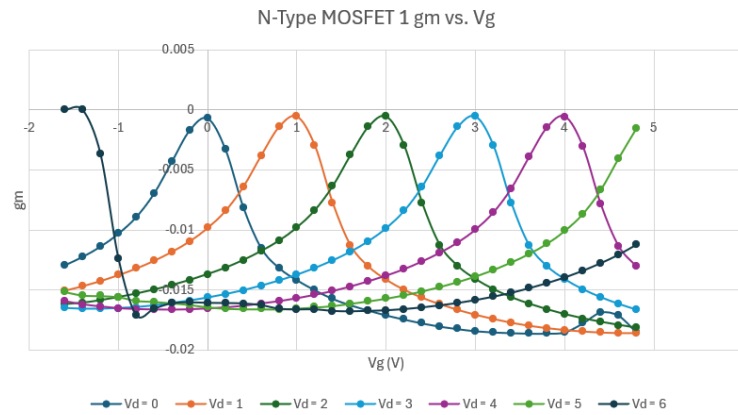
I_D - V_D :



I_D - V_G :



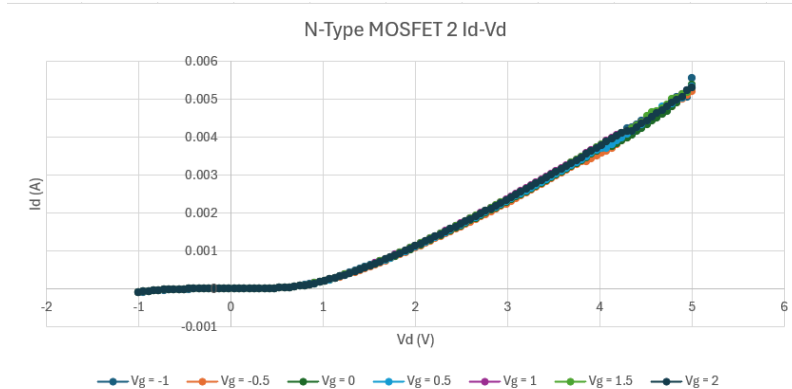
g_m vs. V_G :



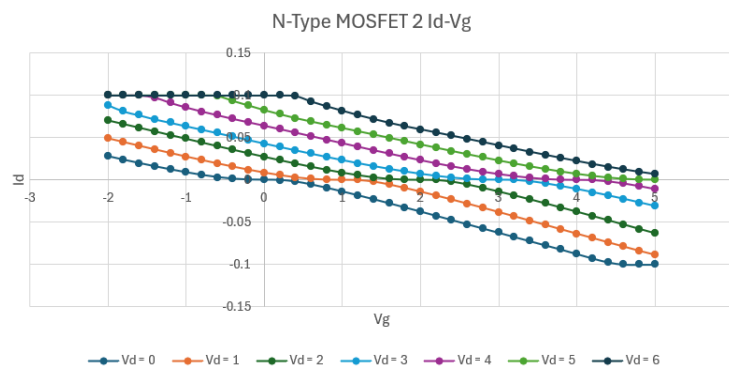
The I_d - V_d curves don't show a clean saturation region and instead keep increasing almost like a resistor, which is likely due to fabrication imperfections like uneven doping or poor channel formation during lithography. If the channel length or the junctions are not well defined, the device will not pinch off correctly. High contact resistance due to imperfect metal connections can also limit current and distort the shape. For the I_d - V_g curves, the fact that current doesn't clearly turn off and appears even at low or negative gate voltages suggests the threshold voltage isn't well controlled. Finally, the g_m vs. V_g plot looks irregular instead of smooth, which is a sign that defects and traps are affecting how the device responds to the gate voltage. These traps can randomly trap and de-trap carriers, causing the transconductance to fluctuate, rather than increase cleanly.

n-type sample, type 2:

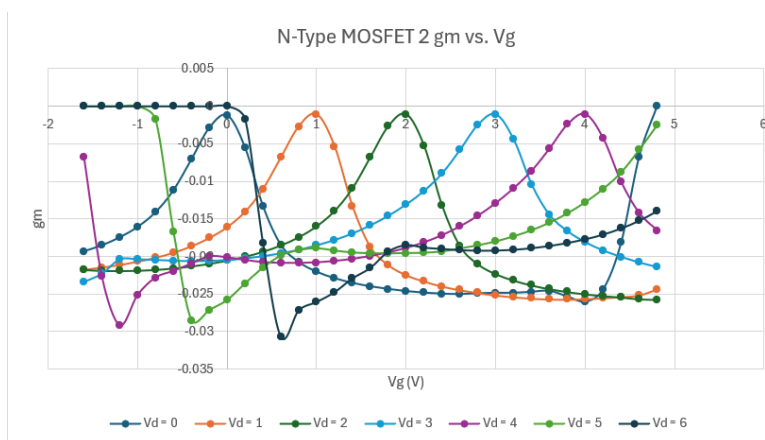
I_D - V_D :



I_D - V_G :



g_m vs. V_G :

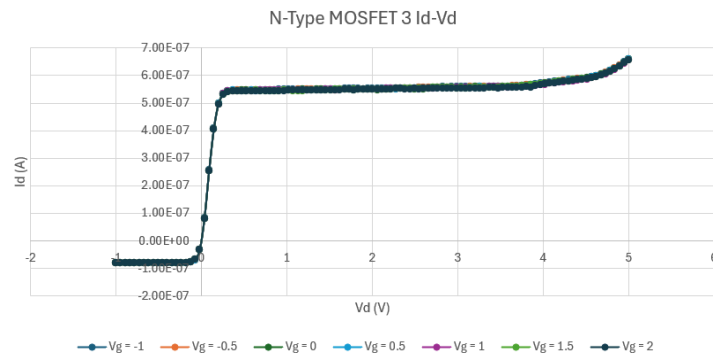


For the I_d - V_d curves the device behaves mostly like a resistor with very little separation of the different V_g curves and no clear saturation which suggests that the gate is not effectively controlling the channel. But processing problems such as bad source/drain formation, high contact resistance or imperfect lithography can dominate even in a thinner oxide, and make this device look like this. The I_d - V_g curves do not show clear turn on and the current does not respond cleanly to the increase of gate voltage. Finally, the g_m vs. V_g plot is very irregular instead of smooth, which is consistent with a device with many

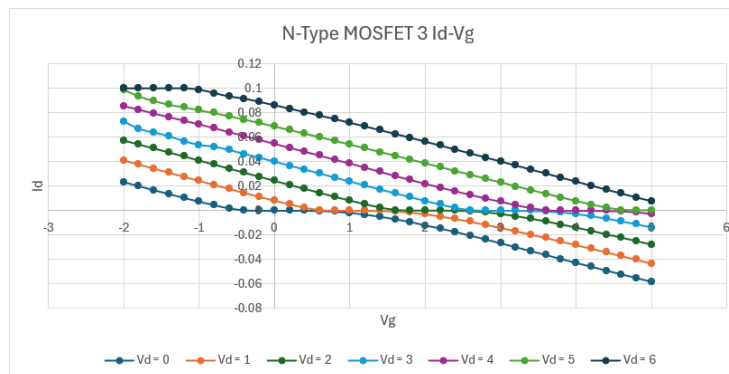
defects and non-uniformities; traps and variations in the channel formation can cause the transconductance to fluctuate rather than increase steadily with gate voltage.

n-type sample, type 3:

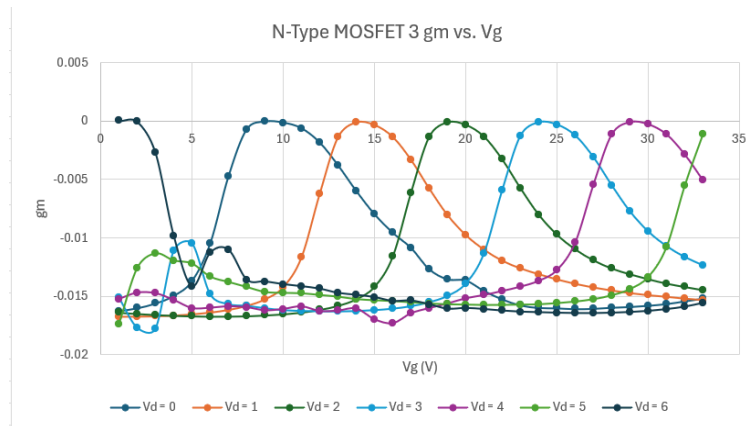
I_D - V_D :



I_D - V_G :



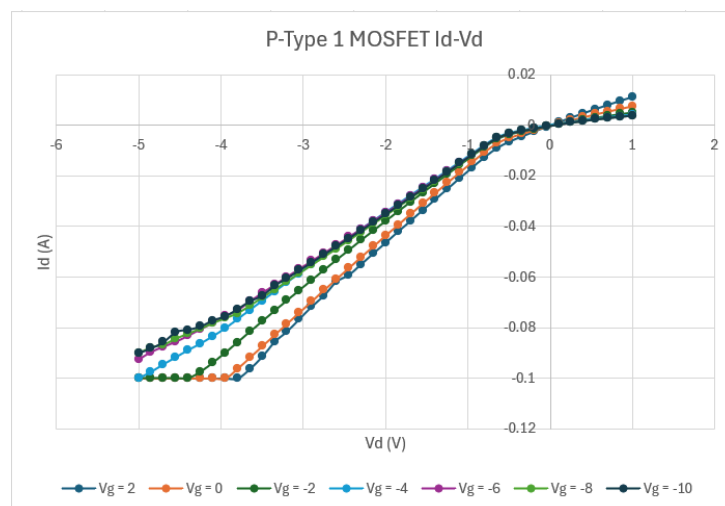
g_m vs. V_G :



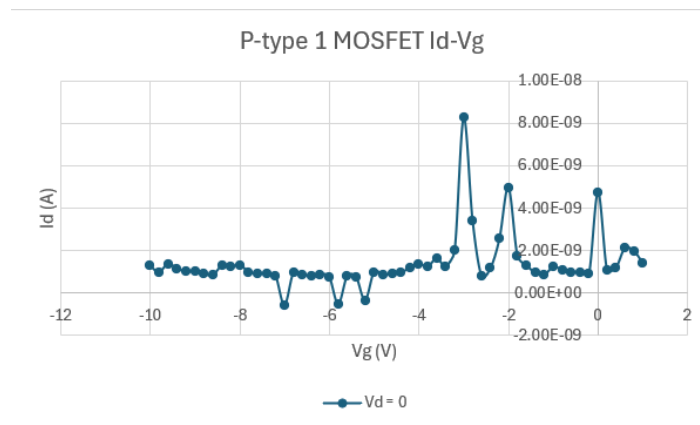
For the I_d - V_d curves, this device does have a much more clear saturation region than the others. However, the total current is very small, which makes sense for a thick oxide as the gate has less control over the channel. The curves still mostly overlap though, which suggests that even when you change V_g , it's not strongly affecting the current, likely due to the thicker oxide and also possible fabrication issues like poor doping or contact resistance. In the I_d - V_g curves, the current still doesn't show a clean turn-on and instead changes in a pretty gradual and somewhat inverted way, which again points to a poorly defined threshold voltage. The g_m vs V_g plot is not smooth and the behavior is not uniform, which means there are defects and non-uniformities that are playing a big role. Things like interface traps or variations in oxide thickness can cause different parts of the device to respond differently to the gate voltage, which can lead to those irregular peaks rather than a clean, consistent increase in transconductance.

p-type sample, type 1:

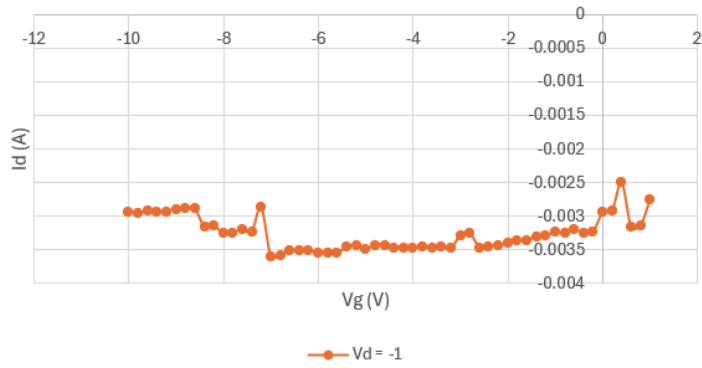
I_D - V_D :



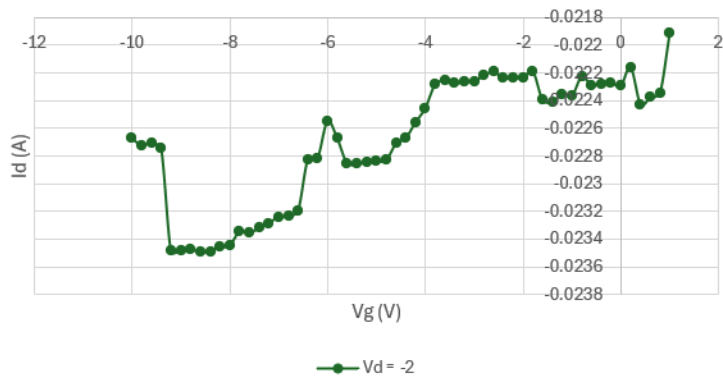
I_D - V_G :



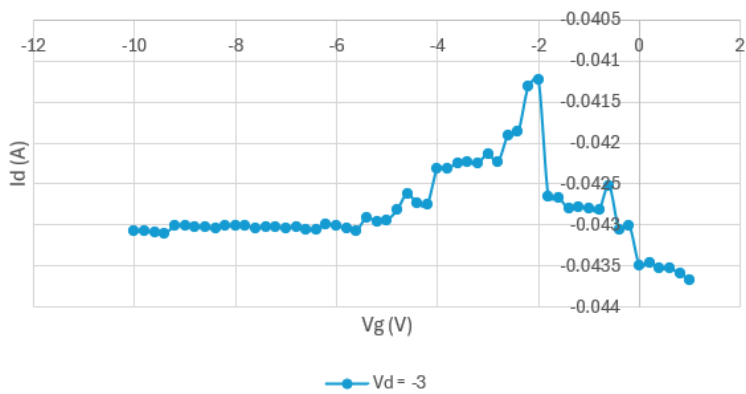
P-type 1 MOSFET Id-Vg

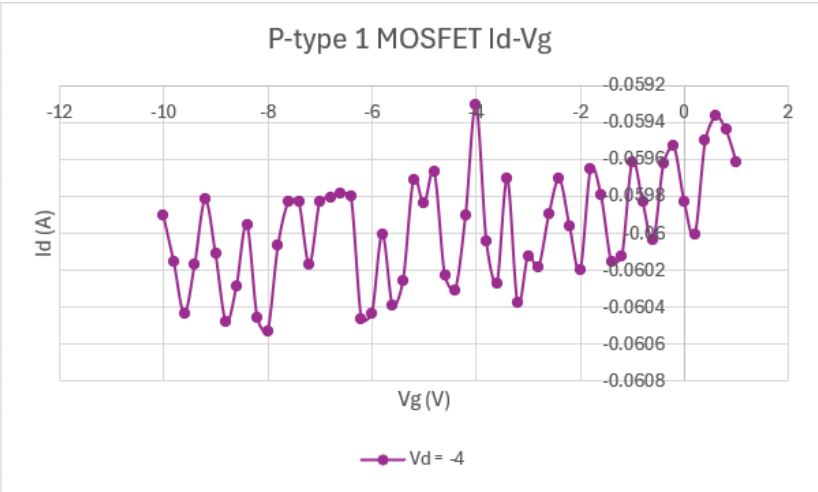


P-type 1 MOSFET Id-Vg

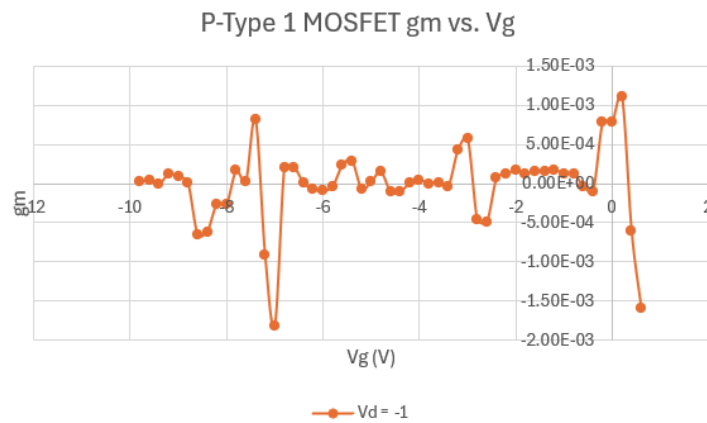
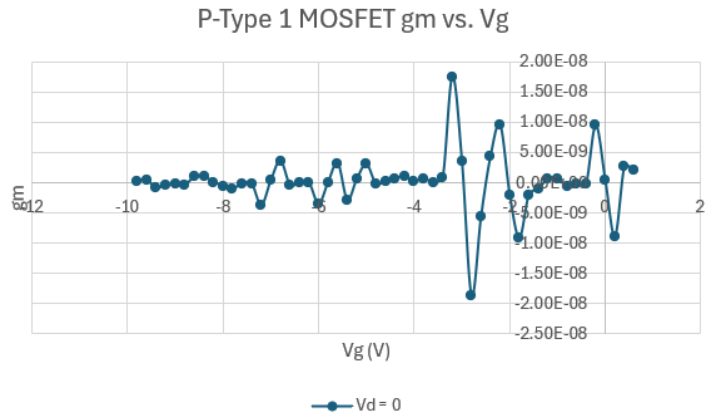


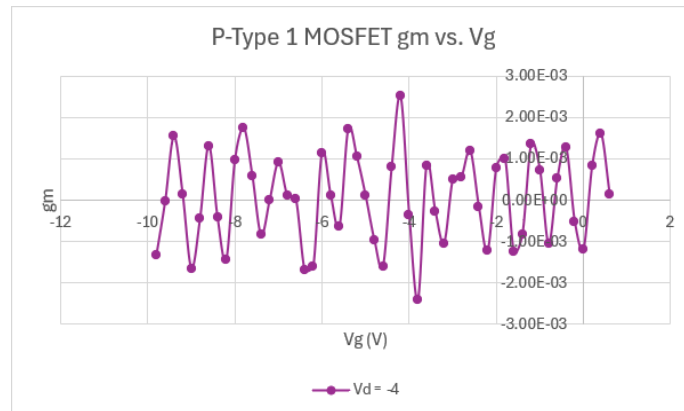
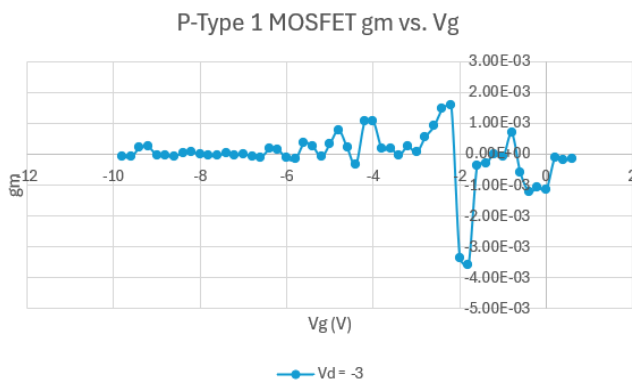
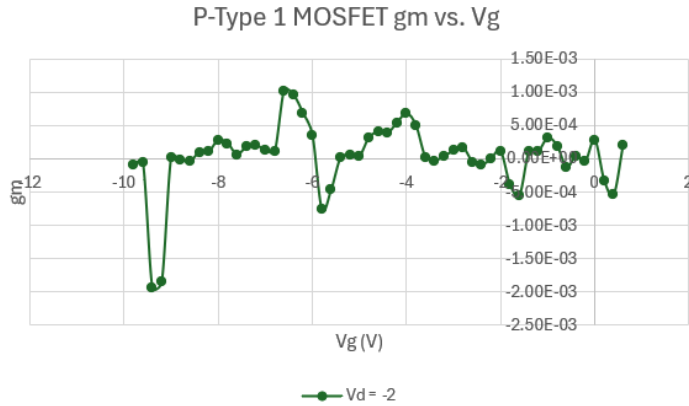
P-type 1 MOSFET Id-Vg





g_m vs. V_G :

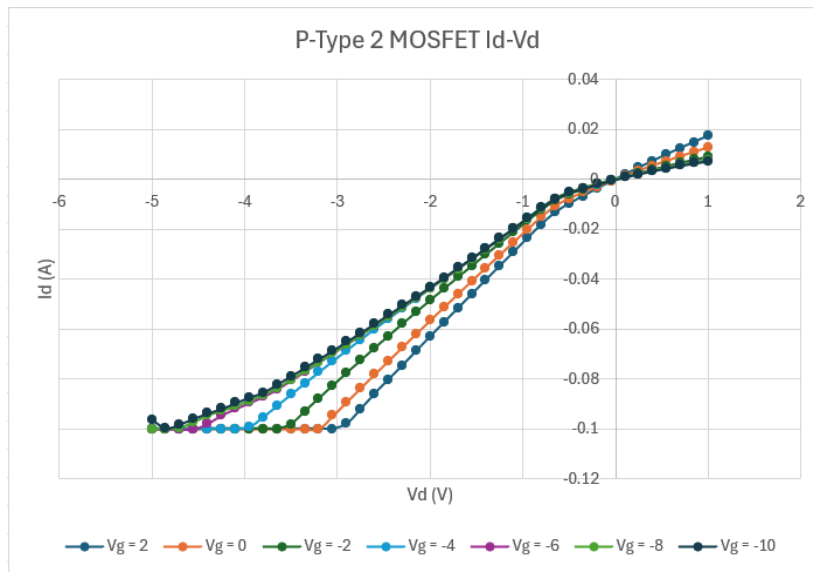




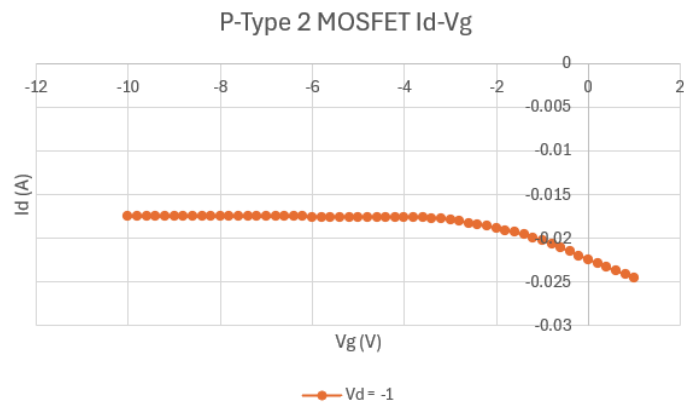
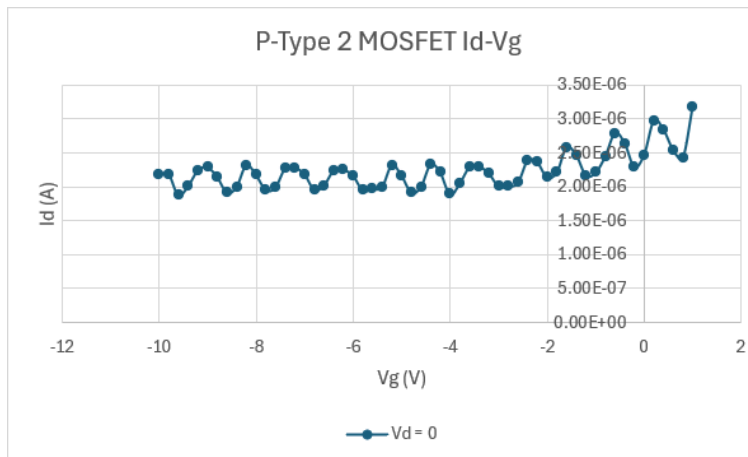
The Type 1 p-type MOSFET does not exhibit clear MOSFET behavior. As seen in the I_D - V_G curves, the drain current shows only weak and inconsistent dependence on gate voltage, with significant noise and no distinct transition between subthreshold and strong inversion regions. The g_m plots are similarly noisy and do not exhibit a clear peak, indicating that transconductance is not well-defined. These observations suggest that the device is dominated by leakage currents and measurement artifacts rather than channel conduction. Possible causes include oxide defects, poor gate contact, or fabrication issues affecting channel formation. As a result, the threshold voltage and subthreshold slope cannot be reliably extracted.

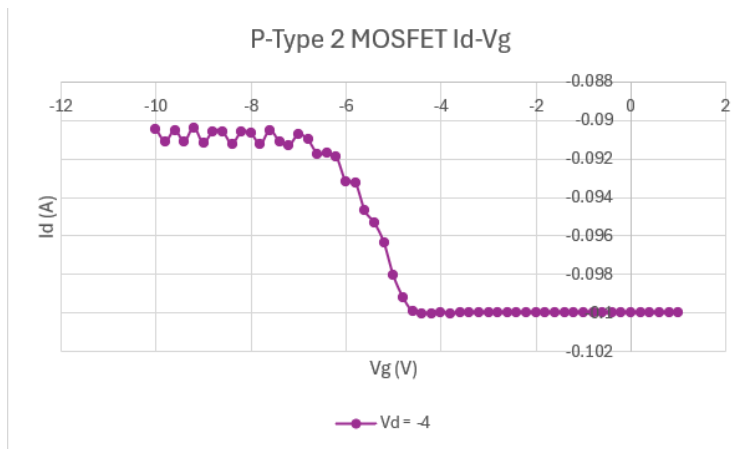
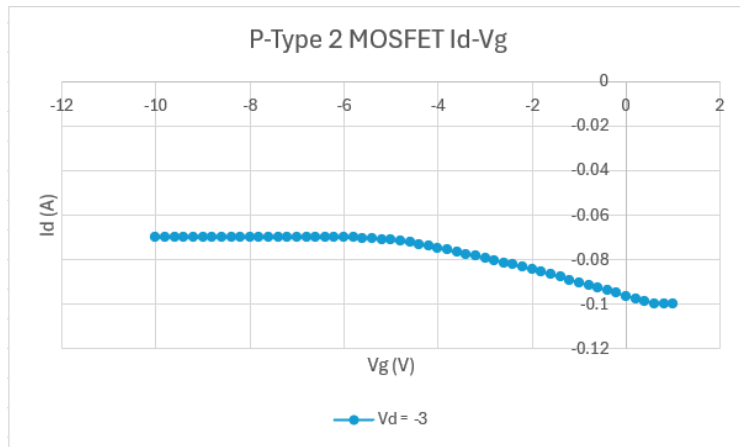
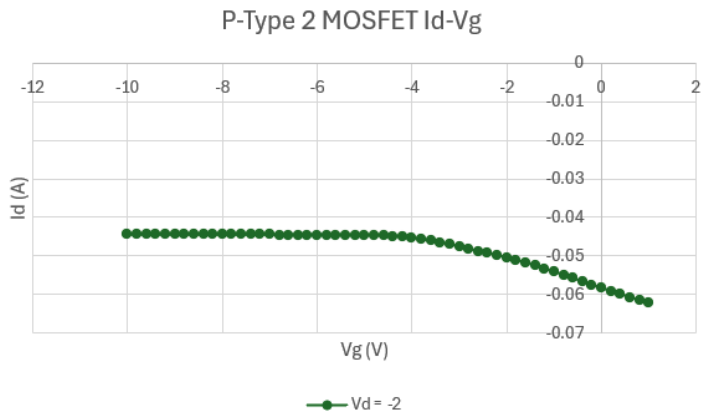
p-type sample, type 2:

I_D - V_D :

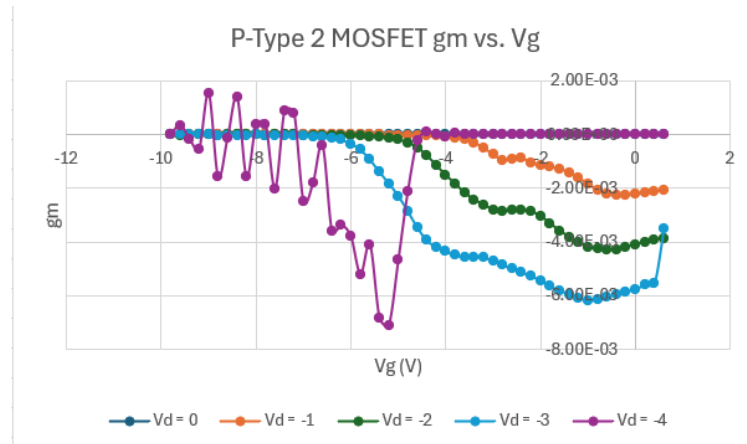


I_D - V_G :





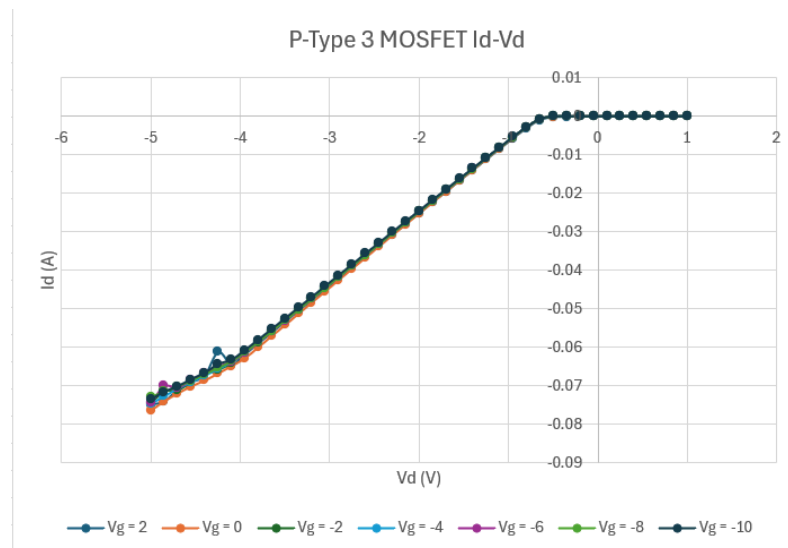
g_m vs. V_G :



The Type 2 p-type MOSFET exhibits clear gate control, as seen in the I_D - V_G curves, where the drain current changes significantly with gate voltage. The most distinct transition occurs for $V_D = -4$ V, where the current begins to increase rapidly around $V_G \approx -4.5$ V to -5 V, which is taken as the threshold voltage. However, the transition is gradual and lacks a sharp exponential region, indicating a degraded subthreshold slope. The g_m plots show a broad and noisy response rather than a sharp peak, further confirming non-ideal behavior. These effects are consistent with short-channel devices, where reduced channel length weakens electrostatic control and increases leakage and variability.

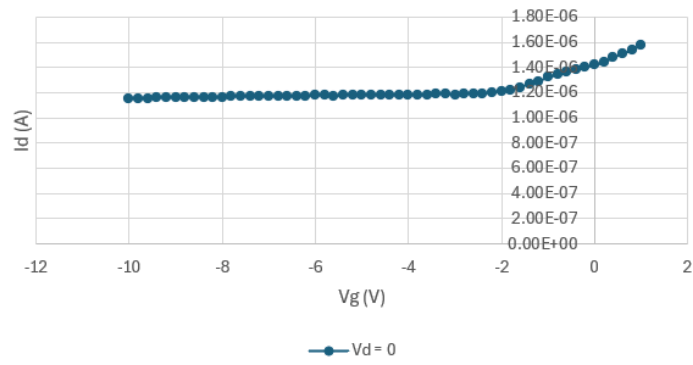
p-type sample, type 3:

I_D - V_D :

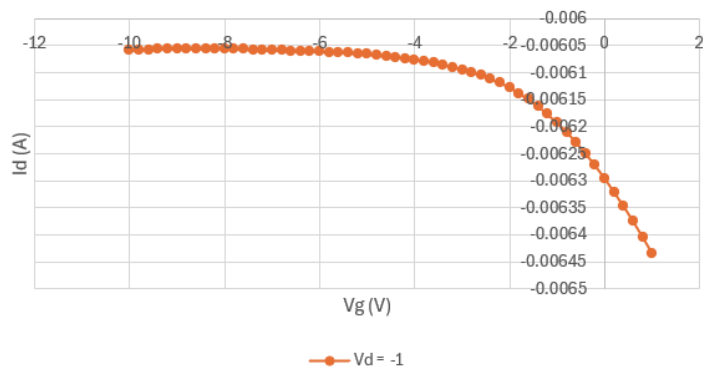


I_D - V_G :

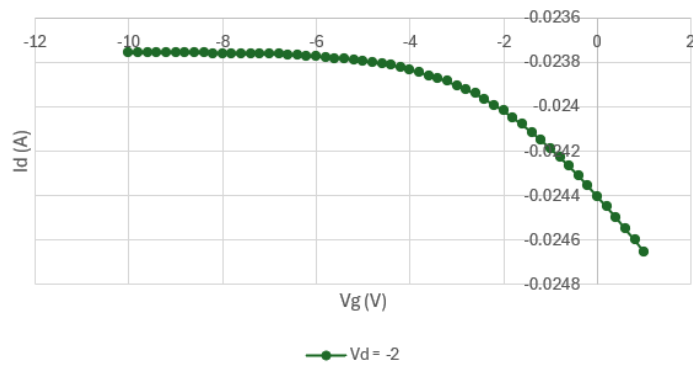
P-Type 3 MOSFET Id-Vg

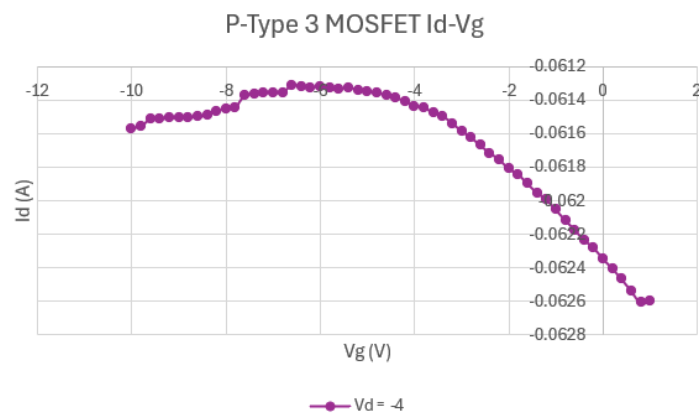
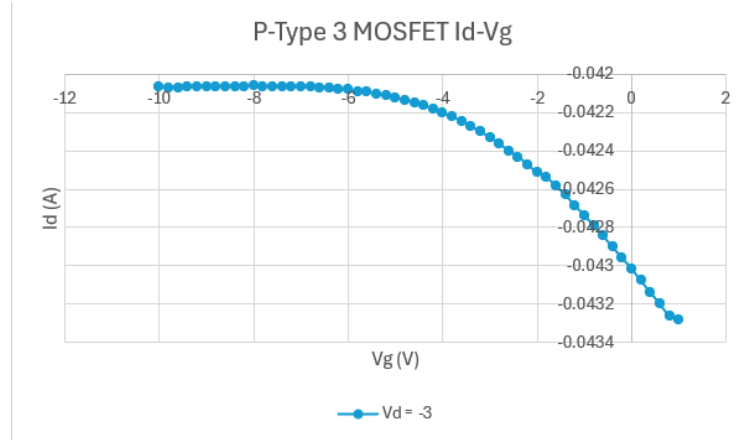


P-Type 3 MOSFET Id-Vg

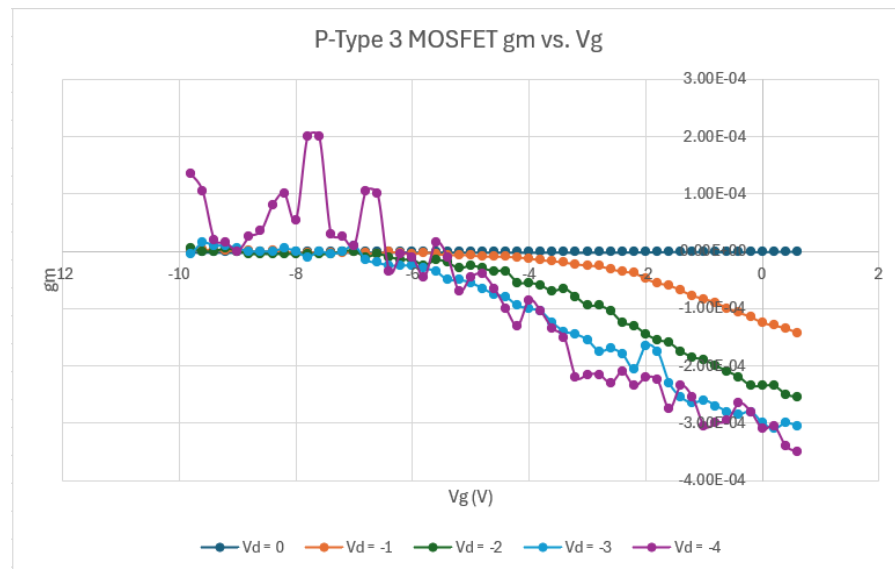


P-Type 3 MOSFET Id-Vg





g_m vs. V_G :



The Type 3 p-type MOSFET shows the most consistent and physically reasonable behavior among the three devices. The I_D - V_G curves exhibit a smooth and monotonic dependence of drain current on gate voltage, with a clear transition region around $V_G \approx$

-3 V, which is taken as the threshold voltage. The g_m plots show a more coherent trend compared to the other devices, indicating more stable transconductance behavior. However, the transition remains gradual, suggesting a non-ideal subthreshold slope. This behavior is consistent with the thick oxide design, which reduces gate capacitance and weakens electrostatic control, leading to a higher threshold voltage and reduced switching sharpness.

■ For both your n- and p-type samples, find the following values for one good MOSFET of each of the 3 kinds. (12 pts)

	n-type sample			p-type sample		
	Type 1	Type 2	Type 3	Type 1	Type 2	Type 3
Threshold voltage (V_{TH})	4.9 V	4.6 V	5.7 V	Unmeasurable	-4.7 V	-3.0 V
Subthreshold slope/swing	unmeasurable	unmeasurable	unmeasurable	Unmeasurable	~500 mV/dec	~250 mV/dec

■ Describe the method you used to find threshold voltage. (1 pt)

The threshold voltage was estimated from the I_D - V_G characteristics by identifying the gate voltage at which the drain current begins to increase significantly, indicating the onset of strong inversion. For devices with clear transitions, this was approximated using the “knee” of the curve, where the slope increases rapidly. In cases where the device exhibited non-ideal or noisy behavior, the threshold voltage could not be reliably extracted.

■ How does the threshold voltage compare to the MOSCAP extracted threshold voltage? (2 pts)

The MOSFET threshold voltages are generally larger in magnitude than those extracted from MOSCAP measurements. This difference arises because MOSCAP measurements reflect ideal surface inversion conditions, while MOSFET threshold voltage is influenced by additional effects such as source/drain resistance, channel length modulation, and interface traps. These non-idealities increase the effective

voltage required to form a conducting channel, leading to a higher magnitude threshold voltage in MOSFET devices.

■ For both your n- and p-type samples, compare the ideal MOSFET equation obtained characteristics to the I-V characteristics in the table above for one good MOSFET of any kind. State which MOSFET you chose. (2 pts)

The p-type Type 3 MOSFET and the n-type type 2 MOSFET were chosen for comparison because they showed the most consistent I_D - V_G behavior. Ideally, a MOSFET in saturation follows:

$$I_D = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{SG} - |V_{TH}|)^2$$

This ideal equation predicts that the drain current should increase approximately quadratically once the gate voltage exceeds threshold, and the transconductance g_m should increase roughly linearly with overdrive voltage. For the p-type Type 3 device, the measured I_D - V_G curves show the expected qualitative trend: the drain current changes smoothly as the gate voltage approaches the threshold region around $V_{TH} \approx -3.0$ V. The g_m plot also shows gate-dependent transconductance, which agrees with the idea that channel charge increases with gate bias. The n-type Type 2 MOSFET was chosen because, despite its non-ideal behavior, it showed the closest qualitative agreement to the expected MOSFET trends compared to the other n-type devices.

However, the measured data does not perfectly follow the ideal square-law model. The transition is gradual instead of sharp, the g_m curve is noisy, and the I_D - V_D curves do not show a clean ideal saturation region. These differences are likely caused by series resistance, contact resistance, leakage current, oxide/interface traps, and nonuniform fabrication. Overall, the p-type Type 3 MOSFET follows the expected MOSFET trend qualitatively, but the measured characteristics are more non-ideal than the

ideal equation predicts. The n-type Type 2 MOSFET, while still non-ideal, showed the closest qualitative agreement to expected MOSFET behavior among the n-type devices.